pubmed_25451397 ? Recognition

The purpose of this study was to address the affects of mood modifying drugs on the transcriptome, in a tissue culture model, using qPCR arrays as a cost effective approach to identifying regulatory networks and pathways that might coordinate the cell response to a specific drug. We addressed the gene expression profile of 90 plus genes associated with human mood disorders using the StellARray? qPCR gene expression system in the human derived SH-SY5Y neuroblastoma cell line. Global Pattern Recognition (GPR) analysis identified a total of 9 genes (DRD3(?), FOS(?), JUN(?), GAD1(??), NRG1(?), PAFAH1B3(?), PER3(?), RELN(?) and RGS4(?)) to be significantly regulated in response to cellular challenge with the mood stabilisers sodium valproate ((?)) and lithium ((?)). Modulation of FOS and JUN highlights the importance of the activator protein 1 (AP-1) transcription factor pathway in the cell response. Enrichment analysis of transcriptional networks relating to this gene set also identified the transcription factor neuron restrictive silencing factor (NRSF) and the oestrogen receptor as an important regulatory mechanism.

Question: Do molecular signatures of mood stabilisers highlight the role of the transcription factor REST/NRSF?

Answer: Cell line models offer a window of what might happen in vivo but have the benefit of being human derived and homogenous with regard to cell type.